Data Structure

data + structure

way to store the data i.e. adding structure our data.

Why important ?

- way we will store our data is going to determine how we solve our problem.

eg:- Bookshelf





a b c d e
f g h i j

student = [ ]

Back and forward button. ⇒ Stack /

Here also talk on other industry use cases.

+ File structure

+ Social media feeds.

Algorithm

Algorithm are step by step instruction or procedures used to manipulate the data stored in the data structure.

Help to solve problems or complete task efficiently using the right data structure.

eg. dictionary



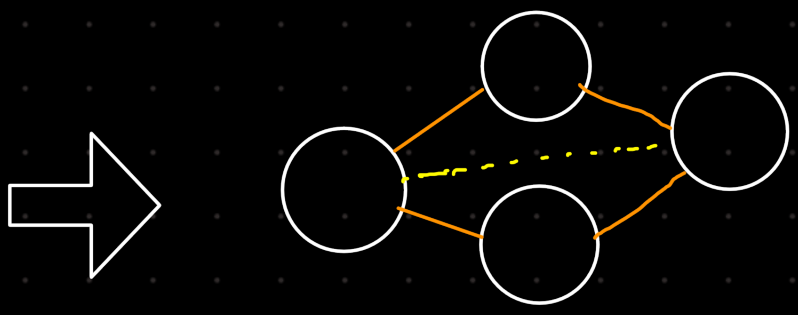
explain how without an algorithm the way of storing as a dictionary / hashmap is still irrelevant.

A → [B, C]

B → [A, D]

C → [A, D]

D → [B, C]



Graph

Now just say how both of the above play a very important role.

in solving example.

Map